AFFINE CASE: $\MATHRM{SUPP}(E) \SUBSET \MATHRM{SUPP}(F)$ WITH F FINITELY PRESENTED FORCES A NONZERO MAP F → E

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 07ZA: sources/stacks-chow/tags.txt, source lines 10140–10140. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00MX: sources/stacks-chow/tags.txt, source lines 839–839. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be an affine scheme. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module. Let $\mathcal{E}$ be a nonzero quasi-coherent $\mathcal{O}_X$-module. If $Supp(\mathcal{E}) \subset Supp(\mathcal{F})$, then there exists a nonzero map $\mathcal{F} \to \mathcal{E}$.*

Proof. Let us translate the statement into algebra. Let A be a ring. Let M be a finitely presented A-module. Let N be a nonzero A-module. Assume $\mathrm{Supp}(N) \subset \mathrm{Supp}(M)$. To show: $\mathrm{Hom}_A(M, N)$ is nonzero. We may assume $N = A/I$ is cyclic (replace N by any nonzero cyclic submodule). Choose a presentation

$$A^{\oplus m} \xrightarrow{T} A^{\oplus n} \to M \to 0$$

Recall that $\mathrm{Supp}(M)$ is cut out by $\mathrm{Fit}_0(M)$ which is the ideal generated by the $n \times n$ minors of the matrix T. See More on Algebra, Lemma 07ZA. The assumption $\mathrm{Supp}(N) \subset \mathrm{Supp}(M)$ now means that the elements of $\mathrm{Fit}_0(M)$ are nilpotent in A/I. Consider the exact sequence

$$0 \to \mathrm{Hom}_A(M, A/I) \to (A/I)^{\oplus n} \xrightarrow{T^t} (A/I)^{\oplus m}$$

We have to show that T^t cannot be injective; we urge the reader to find their own proof of this using the nilpotency of elements of $\mathrm{Fit}_0(M)$ in A/I. Here is our proof. Since $\mathrm{Fit}_0(M)$ is finitely generated, the nilpotency means that the annihilator $J \subset A/I$ of $\mathrm{Fit}_0(M)$ in A/I is nonzero. To show the non-injectivity of T^t we may localize at a prime. Choosing a suitable prime we may assume A is local and J is still nonzero. Then T^t has a nonzero kernel by More on Algebra, Lemma 00MX. □

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